Nakajima-extremal ordinary bielliptic Artin-Schreier curves with full dihedral-times-involution automorphism group

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Allowed standard prerequisites (editorial).

- Artin-Schreier covers, ordinary elliptic curves, p-rank/ramification and bielliptic automorphism rigidity. Original complete prerequisite formulations and bibliography retained.

1 Introduction

Throughout the paper, k is an algebraically closed field of characteristic 2. Let X be a curve of genus $g \geq 2$, let γ be its 2-rank, and let S be a 2-subgroup of $\mathrm{Aut}(X)$. We consider the hypotheses

$$\text{(I)} \quad |S| \geq 8, \qquad |S| > 2(g-1),$$

$$\text{(II)} \quad S \text{ fixes no point of } X.$$

Giulietti and Korchmáros determined the possibilities for (X, S) under these assumptions [6]. A central involution $v \in Z(S)$ is called *inductive* if $X/\langle v\rangle$ has genus at least 2 and the induced group $S/\langle v\rangle$ again satisfies (I) and (II). We recall their classification theorem.

Theorem 1.1 (Giulietti–Korchmáros [6, Theorem 1.2])**.** *Assume that S satisfies* (I) *and* (II)*. Then one of the following cases occurs.*

(i)
$$|S| = 4(g-1), \qquad \gamma = g,$$
and X is bielliptic. Moreover, either

(ia) *S is dihedral and has no inductive central involution; or*

(ib)
$$S = (C_0 \times \langle u\rangle) \rtimes \langle w\rangle,$$
where C_0 is cyclic of order $g-1$, u is a central involution, w is an involution, and $S/\langle u\rangle$ is dihedral. Moreover, S has exactly two central inductive involutions, both contained in $C_0 \times \langle u\rangle$.

(ii) *One has $\gamma = g$ and*
$$|S| = 2g + 2,$$
and $S = D \rtimes \langle w\rangle$, where D is an elementary abelian subgroup of index 2 and w is an involution. If S is abelian, then it is elementary abelian and X is hyperelliptic.

(iii) *Every central involution of S is inductive.*

Proposition 1.2 (Refinement of case (ib) in [6])**.** *In case* (ib)*, let a be the unique involution of C_0. Then a and au are the two inductive central involutions of S, while u is non-inductive; in fact*

$$g(X/\langle u\rangle) = 1.$$

Proof. This is the refinement established in the proof of [6, Theorem 4.3, pp. 277–278]: the involutions a and au are shown there to be inductive, whereas the quotient by u has genus 1. □

Remark 1.3. The published statement of [6, Theorem 1.2(ib)] says that "the two involutions of $E \times \langle u\rangle$" are the unique two central inductive involutions. We denote the cyclic factor in that statement by C_0, to avoid confusion with the elliptic curves used below. Taken literally, the phrase is ambiguous, since $C_0 \times \langle u\rangle$ has three involutions a, u, au, where a is the unique involution of C_0. The proof of [6, Theorem 4.3, pp. 277–278] removes the ambiguity: it shows that a and au are inductive, whereas u is the third central involution and $X/\langle u\rangle$ has genus 1. Accordingly, Theorem 1.1 records the invariant content of the published statement, and Proposition 1.2 identifies the three involutions explicitly.

Proposition 1.4. *Case* (ib) *of Theorem 1.1 does not occur for $g = 3$. Consequently $g - 1 = 2^h \geq 4$ in every instance of case* (ib)*.*

Proof. In case (ib), let a be the unique involution of the cyclic subgroup C_0. By Proposition 1.2, the involution a is inductive. Hence the induced group $S/\langle a\rangle$ satisfies (I). Since $|S| = 4(g-1)$ in case (i),

$$|S/\langle a\rangle| = 2(g-1) \geq 8,$$

so $g \geq 5$. Thus case (ib) cannot occur for $g = 3$.

Finally, S is a 2-group and $|S| = 4(g-1)$, so $g-1$ is a power of 2. Therefore $g - 1 = 2^h$ with $h \geq 2$. □

Remark 1.5. For $g = 3$ the two orders $4(g-1)$ and $2g+2$ are both equal to 8, so cases (i) and (ii) are not separated by the group order alone. The point is that case (ib) contains an inductive central involution, which forces the quotient group to satisfy (I) and hence rules out $g = 3$. Thus every curve in case (ib) has $g - 1 = 2^h \geq 4$, and Theorem 5.6 below will cover all such curves.

We shall be concerned with case (ib). Giulietti and Korchmáros gave isolated examples related to this case; see [6, Sections 6.2 and 6.5]. In Proposition 5.8 below we identify these examples directly with members of our family. In his CoCoA 2015 lectures, Korchmáros posed the following problem [7, Problem 11].

Problem 1.6 (Korchmáros)**.** *Construct an infinite family of curves of type* (ib)*.*

A closely related Artin–Schreier construction over ordinary elliptic curves already appears in [6, Section 5]. Giulietti and Korchmáros start with an ordinary elliptic curve $\bar{\mathcal{X}}$ and, for $n = 2^h \geq 8$, distinguish the subgroup $\bar{\mathcal{X}}[n](k)$ from the nontrivial coset $\bar{\mathcal{X}}[2n](k) \setminus \bar{\mathcal{X}}[n](k)$. Their covers are ramified over the latter set; the translation subgroup of order n lifts non-splitly to a cyclic group of order $2n$, and the resulting extremal group is dihedral; see in particular [6, Proposition 5.12 and Theorem 5.14]. The construction below uses instead the subgroup $E[q](k)$ itself and prescribes equal residues at all its points. This forces the lift of the translation group to split from the deck involution and produces case (ib) rather than the dihedral case (ia). The new ingredients here are this split equal-residue realization, the descent argument giving the converse, and the resulting parametrization of all curves of type (ib).

Artin–Schreier covers and their p-ranks have been studied since the work of Subrao [1]. Large automorphism groups of ordinary curves in characteristic 2 were studied, for example,

by Montanucci–Speziali [2]; their characteristic-2 bound is proved for ordinary curves of even genus, whereas the curves constructed here have odd genus 2^h+1. Our concern is the extremal configuration singled out in [6] and in Problem 1.6. The dihedral construction of [6, Section 5] is closely related to ours, but it has a non-split lift and belongs to case (ia). To the best of our knowledge, the split equal-residue family below, the converse description of all curves of type (ib), and its parameter space do not appear in those works.

We solve this problem by means of Artin–Schreier covers of ordinary elliptic curves. Let $q = 2^h$ and let E/k be ordinary. The group

$$T = E[q](k)$$

is cyclic of order q. We choose a rational function f with simple poles at the points of T and with equal nonzero residues with respect to an invariant differential on E. The translations by T and the elliptic involution lift to the Artin–Schreier cover

$$z^2 + z = f.$$

This gives the following result.

Theorem 1.7. *Let $q = 2^h$ with $h \geq 3$, let E/k be an ordinary elliptic curve, let ω be a nonzero invariant differential on E, and let $\lambda \in k^\times$. Put $T = E[q](k)$. Choose $f \in H^0(E, \mathcal{O}_E(\sum_{P\in T}[P]))$ satisfying*

$$\operatorname{Res}_P(f\omega) = \lambda \qquad (P \in T),$$

and let X be the smooth projective curve with function field $k(E)(z)$, where $z^2 + z = f$. The k-isomorphism class of X is independent of the choice of f; we denote it by $X(E, \omega, \lambda, q)$. This curve has the following properties:

(i) *X is ordinary and bielliptic, with $g(X) = q + 1$;*

(ii) *the deck involution of $X \to E$ is the unique bielliptic involution of X;*

(iii)
$$\operatorname{Aut}(X) \cong \operatorname{Dih}(C_q) \times C_2, \qquad |\operatorname{Aut}(X)| = 4q = 4(g(X) - 1);$$

(iv) *$\operatorname{Aut}(X)$ has no global fixed point on X, and $(X, \operatorname{Aut}(X))$ is of type* (ib)*;*

(v) *there is a cyclic subgroup $C \cong C_q$ acting freely on X, and X/C is an ordinary curve of genus 2.*

The Artin–Schreier construction used in the proof is also defined for $q = 4$; only the assertion on the full automorphism group in Theorem 1.7 requires $h \geq 3$. We use the same notation $X(E, \omega, \lambda, 4)$ below. In particular, curves of type (ib) exist for every

$$g = 2^h + 1, \qquad h \geq 2,$$

where the case $h = 2$ is established in Remark 7.3 below. We prove that every curve in case (ib) arises from this construction. Proposition 1.4 shows that no further genus can occur. For $h \geq 3$ the converse, together with Theorem 1.7, determines the full automorphism group. For $h = 2$ the construction recovers every type (ib) curve of genus 5, although we do not determine its full automorphism group. For every fixed $q \geq 8$, the rigidity result below gives a complete two-parameter classification of isomorphism classes: they are in bijection with $(k^\times)^2$; see Theorem 5.10.

The paper is organized as follows. Section 2 collects the facts on Artin–Schreier covers and ordinary elliptic curves used in the sequel. The construction and the full automorphism group are treated in Section 3. Section 4 gives the short orbits and ramification. Section 5 proves the converse and studies the cyclic quotients. Section 6 deals with the central involutions. Explicit equations are given in Section 7.

2 Preliminaries

We use the following standard facts. If a 2-group H acts on a curve Y and $\Omega_1, \dots, \Omega_r$ are its short orbits, the Deuring–Shafarevich formula (see, for example, [1, 4]) is

$$\gamma(Y) - 1 = |H|\big(\gamma(Y/H) - 1\big) + \sum_{i=1}^{r} \big(|H| - |\Omega_i|\big). \tag{1}$$

Nakajima's bound gives

$$|H| \le 4(\gamma(Y) - 1) \tag{2}$$

when $\gamma(Y) > 1$ [5, Theorem 1]. An ordinary curve is one for which $\gamma(Y) = g(Y)$.

Lemma 2.1. *Let a finite 2-group H act on an ordinary curve Y. Then Y/H is ordinary and the action is weakly ramified. In particular, every point stabilizer H_P is an elementary abelian 2-group. Compare [4].*

Proof. Let $Z = Y/H$. Since H is a 2-group, at every ramified point $H_P^{(0)} = H_P^{(1)} = H_P$, and hence

$$d_P \ge 2(|H_P| - 1).$$

Moreover,

$$\sum_{P \in Y} (|H_P| - 1) = \sum_{\Omega \text{ short}} (|H| - |\Omega|).$$

Riemann–Hurwitz therefore gives

$$2g(Y) - 2 \ge |H|(2g(Z) - 2) + 2 \sum_{\Omega \text{ short}} (|H| - |\Omega|).$$

On the other hand, doubling the Deuring–Shafarevich formula and using $\gamma(Y) = g(Y)$ gives

$$2g(Y) - 2 = 2|H|(\gamma(Z) - 1) + 2 \sum_{\Omega \text{ short}} (|H| - |\Omega|).$$

Since $\gamma(Z) \le g(Z)$, equality must hold throughout. Thus $\gamma(Z) = g(Z)$ and $d_P = 2(|H_P| - 1)$ at every ramified point, so $H_P^{(2)} = 1$. Finally, the standard ramification quotient $H_P^{(1)}/H_P^{(2)}$ is elementary abelian; hence so is H_P. □

If

$$k(X) = k(Y)(z), \qquad z^2 + z = f,$$

and f has a simple pole at $P \in Y$, then the point above P is totally ramified, its lower jump is 1, and its different exponent is 2; see [9, Proposition III.7.8]. We shall also use the Castelnuovo–Severi inequality in the form given in [9, Theorem III.10.3].

Let E/k be an ordinary elliptic curve. For every $h \ge 1$,

$$E[2^h](k) \cong C_{2^h}.$$

A nonzero invariant differential on E has no zero. Since E is ordinary in characteristic 2, one has $j(E) \ne 0$; hence

$$\mathrm{Aut}(E, 0) = \{1, [-1]\}$$

by [8, Theorem III.10.1]. We write

$$\mathrm{Dih}(C_n) = \langle t, w \mid t^n = w^2 = 1,\ wtw = t^{-1} \rangle$$

for the dihedral group of order $2n$.

The following lemma will be used repeatedly.

Lemma 2.2. *Let $T \subset E(k)$ be a finite subgroup of even order q, put*

$$D = \sum_{P \in T} [P],$$

and let ω be a nonzero invariant differential on E. For every $\lambda \in k$ there exists $f \in H^0(E, \mathcal{O}_E(D))$ such that

$$\operatorname{Res}_P(f\omega) = \lambda \qquad (P \in T).$$

If $\lambda \neq 0$, then the poles of f are precisely the points of T and they are simple. Moreover, for every $Q \in T$,

$$\tau_Q^* f - f \in k, \qquad [-1]^* f - f \in k.$$

Proof. Consider

$$\rho : H^0(E, \mathcal{O}_E(D)) \longrightarrow k^T, \qquad f \longmapsto \big(\operatorname{Res}_P(f\omega)\big)_{P \in T}.$$

Since $g(E) = 1$ and $\deg D = q$, Riemann–Roch gives $h^0(E, \mathcal{O}_E(D)) = q$. The kernel of ρ consists of the constants. Indeed, every allowed pole is at most simple; hence, if all residues vanish, every possible simple pole of $f\omega$ is removable. Thus $f\omega$ is regular on E, and since ω is nowhere vanishing, f is constant. Hence $\dim \operatorname{im} \rho = q - 1$. By the residue theorem,

$$\operatorname{im} \rho \subseteq \left\{ (a_P)_{P \in T} : \sum_{P \in T} a_P = 0 \right\},$$

and the two spaces have the same dimension. Since $q = 0$ in k, the constant vector $(\lambda)_{P \in T}$ belongs to this hyperplane. Hence it lies in the image of ρ, which proves the existence of f.

If $Q \in T$, translation by Q preserves ω and permutes T. For every $P \in T$ one has

$$\operatorname{Res}_P\big((\tau_Q^* f)\omega\big) = \operatorname{Res}_{P+Q}(f\omega) = \lambda = \operatorname{Res}_P(f\omega).$$

Thus the possible simple poles of $(\tau_Q^* f - f)\omega$ have zero residue and are removable. Since ω is nowhere vanishing, $\tau_Q^* f - f$ is regular, and hence constant. The same argument applies to $[-1]$: here $[-1](T) = T$ and $[-1]^*\omega = -\omega = \omega$ in characteristic 2, so the residues cancel in exactly the same way. □

3 The construction

3.1 The Artin–Schreier cover

Proof of Theorem 1.7, except for the full automorphism group. Because E is ordinary,

$$E[2^h](k) \cong C_q.$$

Let

$$T = E[2^h](k) = \langle P_0 \rangle.$$

Apply Lemma 2.2 with the prescribed $\lambda \in k^\times$ to obtain $f \in k(E)$ having exactly the points of T as simple poles and satisfying

$$\operatorname{Res}_P(f\omega) = \lambda \qquad (P \in T).$$

Let X be the smooth projective curve with function field

$$k(X) = k(E)(z), \qquad z^2 + z = f. \tag{3}$$

The resulting k-isomorphism class is independent of the choice of f with the prescribed residues. Indeed, if f' is another such function, then $f' - f$ lies in the kernel of the residue map in Lemma 2.2, hence is a constant $c \in k$. Since k is algebraically closed, there is $e \in k$ with $e^2 + e = c$, and the change of Artin–Schreier coordinate $z \mapsto z + e$ identifies the covers defined by f and f'. Thus the notation $X(E, \omega, \lambda, q)$ is unambiguous up to k-isomorphism.

The Artin–Schreier extension is nontrivial. Indeed, a function of the form $r^2 + r + c$ cannot have a pole of odd order, while f has simple poles. Thus $[k(X) : k(E)] = 2$.

At each $P \in T$ the function f has a pole of order one. Standard Artin–Schreier ramification theory shows that the unique point of X above P is totally ramified, with lower jump 1 and different exponent 2; see [9, Proposition III.7.8]. There is no other ramification. Since $g(E) = 1$, Riemann–Hurwitz gives

$$2g(X) - 2 = \sum_{P \in T} 2 = 2q,$$

and hence

$$g(X) = q + 1. \tag{4}$$

Let $\gamma(Y)$ denote the 2-rank of a curve Y. The Deuring–Shafarevich formula for the C_2-cover $X \to E$ gives

$$\gamma(X) - 1 = 2\big(\gamma(E) - 1\big) + q.$$

Since E is ordinary, $\gamma(E) = 1$, and therefore

$$\gamma(X) = q + 1 = g(X).$$

Thus X is ordinary.

Let $\tau = \tau_{P_0}$. By Lemma 2.2,

$$\tau^* f - f = c$$

for some $c \in k$. Choose $\alpha \in k$ with $\alpha^2 + \alpha = c$. Then

$$\widetilde{\tau}(R, z) = (R + P_0, z + \alpha)$$

defines an automorphism of X. Its projection to E has exact order q, and

$$\widetilde{\tau}^q(R, z) = (R, z + q\alpha) = (R, z),$$

so $\widetilde{\tau}$ itself has exact order q.

The deck transformation is

$$u(R, z) = (R, z + 1).$$

Let $\iota = [-1]$ on E. Write $\iota^* f - f = d$ and choose $b \in k$ with $b^2 + b = d$. Then

$$w(R, z) = (-R, z + b)$$

is well defined, since

$$(z + b)^2 + (z + b) = f(R) + d = f(-R),$$

and $w^2 = 1$ because $2b = 0$. Clearly $[u, \widetilde{\tau}] = [u, w] = 1$. Moreover, using $w^{-1} = w$,

$$\begin{aligned} w\widetilde{\tau}w^{-1}(R, z) &= w\widetilde{\tau}(-R, z + b) \\ &= w(-R + P_0, z + b + \alpha) \\ &= (R - P_0, z + \alpha) = \widetilde{\tau}^{-1}(R, z), \end{aligned}$$

where the last equality uses $-\alpha = \alpha$ in characteristic 2. Put $D = \langle \widetilde{\tau}, w \rangle$. The projection $X \to E$ induces a surjection

$$D \longrightarrow \langle \tau_{P_0}, [-1] \rangle \cong \mathrm{Dih}(C_q).$$

The relations above give $|D| \le 2q$, while the image has order $2q$. Hence $D \cong \mathrm{Dih}(C_q)$. The projection is faithful on D, so $D \cap \langle u \rangle = 1$. Since u is central,

$$S := \langle \widetilde{\tau}, w, u \rangle \cong \mathrm{Dih}(C_q) \times C_2,$$

and $|S| = 4q$. The quotient $X/\langle u \rangle$ is E, so X is bielliptic. The subgroup S has no global fixed point, because every nontrivial power of $\widetilde{\tau}$ projects to a nontrivial translation of E and therefore has no fixed point.

In fact the cyclic subgroup

$$C := \langle \widetilde{\tau} \rangle \cong C_q$$

acts freely on X. Put $Y = X/C$. Since $X \to Y$ is an unramified cyclic cover of degree q, Riemann–Hurwitz gives

$$2g(X) - 2 = q\big(2g(Y) - 2\big).$$

Using $g(X) = q + 1$, we obtain $g(Y) = 2$. Likewise, the Deuring–Shafarevich formula for the free C-action gives

$$\gamma(X) - 1 = q\big(\gamma(Y) - 1\big).$$

Since $\gamma(X) = q + 1$, it follows that $\gamma(Y) = 2 = g(Y)$. Hence Y is ordinary. This proves part (v) of the theorem at this stage.

Since $g(X) = q + 1 \ge 9$, the Castelnuovo–Severi inequality implies that X has at most one bielliptic involution: two distinct degree-2 maps to genus-one curves would give

$$g(X) \le 2 \cdot 1 + 2 \cdot 1 + (2 - 1)(2 - 1) = 5;$$

see [9, Theorem III.10.3]. Hence u is the unique bielliptic involution. □

3.2 The full automorphism group

Proposition 3.1. *For a curve $X = X(E, \omega, \lambda, q)$ of Theorem 1.7,*

$$\mathrm{Aut}(X) = S \cong \mathrm{Dih}(C_q) \times C_2.$$

In particular, the group occurring in case (ib) *is the full automorphism group of X.*

Proof. The unique bielliptic involution u is central in $\mathrm{Aut}(X)$: for any $\alpha \in \mathrm{Aut}(X)$, the conjugate $\alpha u \alpha^{-1}$ is again a bielliptic involution, hence equals u. Therefore every automorphism of X descends through

$$\pi : X \longrightarrow E = X/\langle u \rangle.$$

The descended automorphism preserves the branch locus of π, which is precisely T. The kernel of the descent homomorphism is the deck group $\langle u \rangle$. Thus

$$\mathrm{Aut}(X)/\langle u \rangle \hookrightarrow \mathrm{Aut}(E, T), \tag{5}$$

where $\mathrm{Aut}(E, T)$ denotes the automorphisms of the curve E preserving T setwise.

Every automorphism α of the genus-one curve E can be written uniquely in the form

$$\alpha = \tau_Q \circ \varphi, \qquad Q = \alpha(0), \quad \varphi \in \mathrm{Aut}(E, 0).$$

If α preserves T setwise, then $Q = \alpha(0) \in T$. Since E is ordinary in characteristic 2, one has $j(E) \neq 0$ and

$$\mathrm{Aut}(E, 0) = \{1, [-1]\}$$

by [8, Theorem III.10.1]; both elements preserve T. Conversely every translation by a point of T, and $[-1]$, preserve T. Hence

$$\mathrm{Aut}(E, T) = T \rtimes \langle [-1] \rangle \cong \mathrm{Dih}(C_q).$$

The subgroup $S/\langle u\rangle$ already maps isomorphically onto this whole group. The injection (5) therefore forces

$$\mathrm{Aut}(X)/\langle u\rangle = S/\langle u\rangle.$$

Since $S \subseteq \mathrm{Aut}(X)$ and both contain u, it follows that $\mathrm{Aut}(X) = S$. □

Proposition 3.1 completes the proof of Theorem 1.7. In particular,

$$|\mathrm{Aut}(X)| = 4q = 4(g(X) - 1).$$

Since $q \geq 8$, one has $|\mathrm{Aut}(X)| \geq 8$ and $|\mathrm{Aut}(X)| = 4q > 2q = 2(g(X) - 1)$, so hypothesis (I) holds; hypothesis (II) was proved above. Hence the classification theorem applies. Case (ii) is impossible because $|\mathrm{Aut}(X)| = 4(g - 1)$ whereas case (ii) has order $2g + 2$; these are equal only for $g = 3$, while here $g \geq 9$. Case (iii) is impossible because the central involution u has quotient E of genus 1 and hence is not inductive. Finally the group is not dihedral: its center has order 4, whereas a dihedral group of the same order $4q$ has center of order 2. Therefore case (ia) is excluded. Thus $(X, \mathrm{Aut}(X))$ is exactly of type (ib).

Corollary 3.2. *Case* (ib) *occurs exactly in the genera*

$$g = 2^h + 1, \qquad h \geq 2.$$

In particular, there are infinitely many genera for which curves of type (ib) *exist.*

Proof. Necessity is Proposition 1.4. For $h \geq 3$ existence is Theorem 1.7, and the case $h = 2$ is given by Remark 7.3. □

4 Ramification and the group action

4.1 Short orbits and ramification

X is ordinary, so $\gamma(X) = g(X) = q + 1$, and therefore

$$|\mathrm{Aut}(X)| = 4q = 4(\gamma(X) - 1).$$

Thus the full automorphism group is itself a 2-group attaining Nakajima's bound. The orbit structure can also be read off completely.

Proposition 4.1. *Let $G = \mathrm{Aut}(X)$. Then*

$$X/G \cong \mathbf{P}^1.$$

Moreover, G has exactly two short orbits Ω_1, Ω_2, with

$$|\Omega_1| = 2q, \qquad |\Omega_2| = q.$$

The second orbit is the ramification locus of the Artin–Schreier map $\pi : X \to E$; its point stabilizers are isomorphic to $C_2 \times C_2$. The stabilizer of a point in the first short orbit is cyclic of order 2. All remaining G-orbits are long.

Proof. Write $T = \langle P_0\rangle \cong C_q$. Since u is central and $X/\langle u\rangle = E$, we have

$$X/G \cong E/(T \rtimes \langle[-1]\rangle).$$

The quotient E/T is again an elliptic curve, and $[-1]$ descends to its elliptic involution. Hence

$$E/(T \rtimes \langle[-1]\rangle) \cong (E/T)/\langle[-1]\rangle \cong \mathbf{P}^1.$$

Let Ω_2 be the set of ramification points of π. There is one point of X above each point of T, so $|\Omega_2| = q$. The translation subgroup acts transitively on this set; hence Ω_2 is a G-orbit. Its stabilizer has order $|G|/q = 4$. Since the cyclic translation subgroup acts freely on X, the stabilizer meets it trivially; it contains the deck involution u. If w_P denotes the reflection fixing the corresponding point $P \in T$ (its projection to E is $\tau_{2P} \circ [-1]$), then the stabilizer is $\{1, u, w_P, uw_P\}$. Since u and w_P are commuting involutions, this stabilizer is a Klein four group.

Now apply the Deuring–Shafarevich formula to the full 2-group G. Since X is ordinary and X/G is rational,

$$q = \gamma(X) - 1 = -4q + \sum_{\Omega \text{ short}} (4q - |\Omega|).$$

The Deuring–Shafarevich identity therefore requires the total short-orbit contribution to be $5q$. The known orbit Ω_2 contributes $3q$, so exactly $2q$ remains. On the other hand, because the cyclic subgroup C_q acts freely, every point stabilizer injects into G/C_q, which has order 4. Thus every further short orbit has length q or $2q$. A further orbit of length q would already contribute $3q$, which is too much. The remaining contribution is exactly $2q$, so there is precisely one further short orbit, it has length $2q$, and its stabilizer has order 2. □

Corollary 4.2. *The action of G on X is weakly ramified. More explicitly, if $P \in \Omega_2$, then*

$$G_P^{(0)} = G_P^{(1)} \cong C_2 \times C_2, \qquad G_P^{(2)} = 1,$$

and the different exponent at P is 6. If $P \in \Omega_1$, then

$$G_P^{(0)} = G_P^{(1)} \cong C_2, \qquad G_P^{(2)} = 1,$$

and the different exponent at P is 2. Consequently the total different of $X \to X/G$ has degree $10q$.

Proof. By Lemma 2.1, the action is weakly ramified. Hence $G_P^{(2)} = 1$, while for a ramified point of a 2-group action one has $G_P^{(0)} = G_P^{(1)} = G_P$. Proposition 4.1 therefore gives local different exponent

$$2(|G_P| - 1),$$

which is 6 on Ω_2 and 2 on Ω_1. Thus

$$\deg \mathrm{Diff}(X/(X/G)) = q \cdot 6 + 2q \cdot 2 = 10q.$$

This also checks Riemann–Hurwitz:

$$2g(X) - 2 = 2q = -2|G| + 10q = -8q + 10q.$$

□

5 The converse and cyclic quotients

5.1 Isomorphisms in the family

The parameter occurring in the construction is intrinsic.

Proposition 5.1. *Let $q = 2^h$ with $h \geq 3$. Let E_i/k be ordinary elliptic curves and let ω_i be nonzero invariant differentials. Then*

$$X(E_1, \omega_1, \lambda_1, q) \cong X(E_2, \omega_2, \lambda_2, q)$$

if and only if there is an isomorphism of elliptic curves $\varphi : E_1 \to E_2$ such that

$$\varphi^*(\lambda_2^{-1}\omega_2) = \lambda_1^{-1}\omega_1.$$

In particular,

$$X(E, \omega, \lambda, q) \cong X(E, \lambda^{-1}\omega, 1, q).$$

Thus, after normalizing the residue to 1*, the curves are parametrized by isomorphism classes of pairs (E, η), where E is ordinary and $0 \neq \eta \in H^0(E, \Omega^1_E)$.*

Proof. The last displayed isomorphism follows at once from

$$\mathrm{Res}_P(f\omega) = \lambda \quad \Longleftrightarrow \quad \mathrm{Res}_P(f\,\lambda^{-1}\omega) = 1.$$

Suppose that

$$\Phi : X(E_1, \eta_1, 1, q) \longrightarrow X(E_2, \eta_2, 1, q)$$

is an isomorphism. The bielliptic involution is unique, so Φ descends to an isomorphism $\psi : E_1 \to E_2$ carrying $E_1[q](k)$ onto $E_2[q](k)$. In particular $\psi(0) \in E_2[q](k)$. Translation by $-\psi(0)$ preserves the branch set and lifts to the second cover. After composing Φ with such a lift, we may therefore assume that the descended map preserves the origins; denote it by $\varphi : E_1 \to E_2$. Write

$$\varphi^*\eta_2 = c\eta_1, \qquad c \in k^\times.$$

Let f_i be Artin–Schreier functions defining the two covers. An isomorphism of quadratic Artin–Schreier extensions over φ sends an Artin–Schreier generator to the other generator plus an element of the base field. Hence

$$\varphi^* f_2 - f_1 = r^2 + r$$

for some $r \in k(E_1)$. The left-hand side has at most simple poles, whereas, if r has a pole of order $m > 0$, then $r^2 + r$ has a pole of order $2m$. Thus r has no pole and is constant. For $P \in E_1[q](k)$ we then have

$$\mathrm{Res}_P(\varphi^* f_2\, \eta_1) = c^{-1}\,\mathrm{Res}_{\varphi(P)}(f_2\eta_2) = c^{-1}, \qquad \mathrm{Res}_P(f_1\eta_1) = 1.$$

Since the difference has zero residue, $c^{-1} = 1$, and therefore $c = 1$.

Conversely, suppose that $\varphi : E_1 \to E_2$ preserves origins and satisfies $\varphi^*\eta_2 = \eta_1$. Then φ carries $E_1[q](k)$ onto $E_2[q](k)$, and $\varphi^* f_2$ and f_1 have the same residues at every point of $E_1[q](k)$. By Lemma 2.2 their difference is constant; since k is algebraically closed, that constant is of the form $e^2 + e$. The change of Artin–Schreier coordinate $z \mapsto z + e$ gives the required isomorphism. □

Corollary 5.2. *For fixed $q \geq 8$, after normalizing the residue to* 1*, the isomorphism classes obtained from the construction are in bijection with the isomorphism classes of pairs (E, η), where E is ordinary elliptic and $0 \neq \eta \in H^0(E, \Omega^1_E)$.*

5.2 Artin–Schreier descent and the converse

The residue construction is not merely a source of examples. It recovers every curve occurring in case (ib).

Lemma 5.3 (Local Artin–Schreier residue criterion)**.** *Let $K = k((t))$, where k is algebraically closed of characteristic* 2*, and let ϑ be a regular differential which is nonzero at $t = 0$. Let $g_1, g_2 \in K$ have at most a simple pole. If*

$$\mathrm{Res}(g_1\vartheta) = \mathrm{Res}(g_2\vartheta),$$

then $g_1 - g_2 \in k[[t]]$. In particular the Artin–Schreier equation

$$y^2 + y = g_1 - g_2$$

is unramified at $t = 0$.

Proof. Write $\vartheta = v(t)\,dt$ with $v(0) \neq 0$. The polar part of $g_1 - g_2$ is c/t for some $c \in k$, and the residue of $(g_1 - g_2)\vartheta$ is $cv(0)$. Hence $c = 0$. Thus $g_1 - g_2 \in k[[t]]$, and the last assertion follows since the derivative of $y^2 + y - (g_1 - g_2)$ with respect to y is 1. □

Artin–Schreier descent lemma 5.4. *Let k be an algebraically closed field of characteristic 2, let E/k be an ordinary elliptic curve, and let*

$$T = E[q](k) \cong C_q, \qquad q = 2^h, \quad h \geq 1.$$

Write $\phi : E \to F := E/T$ for the quotient isogeny. Let $\pi : Y \to E$ be a connected Artin–Schreier double cover with deck involution u. Suppose that the branch locus of π is exactly T, with lower jump 1 at every branch point, and that the translations by T lift to a subgroup $\widetilde{T} \leq \mathrm{Aut}(Y)$ such that

$$\widetilde{T} \cong T, \qquad \widetilde{T} \cap \langle u \rangle = 1, \qquad [\widetilde{T}, u] = 1.$$

Then, for every nonzero invariant differential ω on E, there is a unique $\lambda \in k^\times$ such that

$$Y \cong X(E, \omega, \lambda, q)$$

as double covers of E.

Proof. Put $Z = Y/\widetilde{T}$. Since $\widetilde{T}$ commutes with u and has trivial intersection with $\langle u \rangle$, the subgroup $\widetilde{T} \times \langle u \rangle$ has order $2q$, and its fixed field is $k(F)$. Inside $k(Y)$ we have

$$k(Z) = k(Y)^{\widetilde{T}}, \qquad k(E) = k(Y)^{\langle u \rangle}.$$

Moreover,

$$k(Z) \cap k(E) = k(Y)^{\langle \widetilde{T}, u \rangle} = k(F).$$

Consequently $k(Y) = k(Z)k(E)$ over $k(F)$, so Y is the normalization of $Z \times_F E$.

The isogeny $\phi : E \to F$ is finite *étale*: its kernel is the reduced constant subgroup scheme determined by the q distinct points of T. The pullback of $Z \to F$ along ϕ is $Y \to E$. Since the latter is ramified exactly at the fibre $T = \phi^{-1}(O_F)$, the cover $Z \to F$ is ramified exactly at O_F. Ramification groups are unchanged by unramified extension of complete local fields; hence the lower jump of Z/F at O_F is again 1.

Let θ be the unique invariant differential on F satisfying $\phi^*\theta = \omega$. Choose a reduced local Artin–Schreier equation for Z/F at O_F,

$$y^2 + y = \frac{a}{t} + b(t), \qquad a \neq 0, \quad b(t) \in k[[t]],$$

and define

$$\lambda = \mathrm{Res}_{O_F}\left(\frac{a}{t}\theta\right).$$

Because the pole is simple, changing the reduced Artin–Schreier representative by $r^2 + r$ cannot alter its simple polar coefficient; thus λ is independent of the chosen reduced equation and is nonzero.

Let $X_\lambda = X(E, \omega, \lambda, q)$, and let $Z_\lambda = X_\lambda/C_\lambda$, where $C_\lambda \cong T$ is the cyclic subgroup lifting the translations by T. By the same quotient and unramified-base-change argument, $Z_\lambda \to F$ is ramified only at O_F, has lower jump 1, and has local residue λ there. Let $\xi, \xi_\lambda \in k(F)/\wp(k(F))$ be the global Artin–Schreier classes of Z/F and Z_λ/F. At O_F, choose reduced representatives of their images in the completed local field. They have at most a simple pole and the same residue with respect to θ; hence Lemma 5.3 shows that the local class of $\xi - \xi_\lambda$ is unramified at O_F. At every other place both local classes are unramified, and so is their difference. Thus $\xi - \xi_\lambda$ is an everywhere-unramified quadratic Artin–Schreier class on F.

Because F is an ordinary elliptic curve, its 2-rank is 1; equivalently, the $\mathbf{F}_2$-vector space of *étale* C_2-torsors on F has dimension 1. Hence F has exactly one nontrivial connected *étale* C_2-cover. Let $T' \leq T$ be the subgroup of order $q/2$ (for $q = 2$ take $T' = 0$). The separable isogeny

$$E/T' \longrightarrow E/T = F$$

is a nontrivial *étale* double cover, and therefore represents this unique nonzero unramified Artin–Schreier class. After pullback along $\phi : E \to F$ it becomes trivial: the factorization

$$E \longrightarrow E/T' \longrightarrow F$$

defines a section of the pulled-back C_2-torsor. Thus the possible unramified difference between Z and Z_λ disappears after base change to E. Since the respective pullbacks are Y and X_λ, we obtain

$$Y \cong X(E, \omega, \lambda, q)$$

as double covers of E.

Finally suppose that the same double cover of E were obtained from parameters λ and λ'. If f, f' are corresponding Artin–Schreier functions, an isomorphism over E gives $f' - f = r^2 + r$ for some $r \in k(E)$. The left-hand side has at most simple poles. Therefore r has no pole, so r is constant, and $f' - f$ has zero residue at every point of T. Hence $\lambda' = \lambda$. This proves uniqueness. □

Remark 5.5. The condition $\widetilde{T} \cap \langle u \rangle = 1$ in the descent lemma is essential. Without it, the inverse image of the translation group need not split as $C_q \times C_2$: the non-split possibility is cyclic of order $2q$, with a lift $\widetilde{t}$ satisfying $\widetilde{t}^q = u$. This is the extension pattern produced by the nontrivial *étale* twist in the descent picture. It is the mechanism exhibited by the dihedral case-(ia) quotient X^+ in Theorem 6.2 below.

Theorem 5.6 (Converse to the construction)**.** *Let Y/k be a smooth projective curve over an algebraically closed field k of characteristic 2, and suppose that Y admits a 2-subgroup $S \leq \mathrm{Aut}(Y)$ satisfying* (I) *and* (II) *such that (Y, S) is in case* (ib) *of Theorem 1.1. Put $q = g(Y) - 1$. Then $q = 2^h$ for some $h \geq 2$, and there are an ordinary elliptic curve E/k and a nonzero invariant differential η on E such that*

$$Y \cong X(E, \eta, 1, q).$$

If $h \geq 3$, then

$$S = \mathrm{Aut}(Y) \cong \mathrm{Dih}(C_q) \times C_2.$$

If $h = 2$, then $S \cong \mathrm{Dih}(C_4) \times C_2$ and S is a Sylow 2-subgroup of $\mathrm{Aut}(Y)$.

Proof. By Proposition 1.4, case (ib) forces $q = g(Y) - 1 = 2^h$ with $h \geq 2$. Write, as in case (ib) of [6, Theorem 1.2],

$$S = (C \times \langle u \rangle) \rtimes \langle w \rangle, \qquad C \cong C_q,$$

where $S/\langle u \rangle$ is dihedral of order $2q$. By Proposition 1.2, u is the central non-inductive involution. The classification also gives

$$\gamma(Y) = g(Y) = q + 1.$$

Put

$$Z := Y/\langle u \rangle$$

and write $g_0 = g(Z)$ and $\gamma_0 = \gamma(Z)$. Let r be the number of fixed points of u. The Deuring–Shafarevich formula gives

$$q = 2(\gamma_0 - 1) + r. \tag{6}$$

On the other hand, Riemann–Hurwitz gives

$$2q = 2(2g_0 - 2) + \deg \mathrm{Diff}(Y/Z).$$

Every fixed point of an involution in characteristic 2 contributes at least 2 to the different. Hence, using (6),

$$\begin{aligned} 2q &\geq 4g_0 - 4 + 2r \\ &= 2q + 4(g_0 - \gamma_0). \end{aligned}$$

Since always $\gamma_0 \leq g_0$, equality must hold throughout. Thus Z is ordinary and every ramified point of $Y \to Z$ has different exponent 2, equivalently lower jump 1.

We claim that $g_0 = 1$. First, $g_0 \neq 0$. Indeed, $S/\langle u \rangle$ acts faithfully on Z, and it contains the image of C, a cyclic group of order $q \geq 4$. But in characteristic 2 every nontrivial element of 2-power order in $\mathrm{PGL}_2(k) = \mathrm{Aut}(\mathbf{P}^1)$ is unipotent, and hence has order 2. Thus a cyclic subgroup of order $q \geq 4$ cannot act faithfully on a rational curve.

Suppose that $g_0 \geq 2$. Since Z is ordinary, (6) gives

$$2(g_0 - 1) = q - r \leq q,$$

and therefore the induced group $S/\langle u \rangle$, of order $2q$, still satisfies hypothesis (I). Since u is non-inductive, hypothesis (II) must fail; hence $S/\langle u \rangle$ fixes a point of Z. By Lemma 2.1, its point stabilizer is elementary abelian. This is impossible because it contains the image of a generator of C, which has order $q \geq 4$. Hence $g_0 = 1$.

It follows from (6) that $\gamma_0 = 1$ and $r = q$. Thus Z is an ordinary elliptic curve, the cover $Y \to Z$ is branched at exactly q points, and the lower jump is 1 at every branch point.

The subgroup C descends faithfully to Z. Choose temporarily an origin $O \in Z(k)$ and a generator $c \in C$. Since Z is ordinary in characteristic 2, $\mathrm{Aut}(Z, O) = \{1, [-1]\}$. Writing the descended automorphism as a translation followed by an element of $\mathrm{Aut}(Z, O)$, its linear part cannot be $[-1]$, because every automorphism of the form $\tau_Q \circ [-1]$ is an involution. The image of c, which has order $q \geq 4$, is therefore translation by a point of exact order q. Hence C acts on Z as translations by a cyclic subgroup T of order q.

Let $B \subset Z$ be the branch locus. It is C-invariant. Translation by T is free on Z, and $|B| = |T| = q$, so $B = Q + T$ for some $Q \in Z(k)$. Use Q as the origin of the genus-one curve Z, and denote the resulting elliptic curve by E. If $+$ denotes the old group law, the new group law is

$$P \oplus P' = P + P' - Q.$$

Thus the old translation $P \mapsto P + t$ is translation, for the new group law, by the point $Q + t$, and

$$(Q + t_1) \oplus (Q + t_2) = Q + (t_1 + t_2).$$

Consequently $B = Q + T$ is a cyclic subgroup of order q in the new group law, and every element of B is killed by q. Since an ordinary elliptic curve in characteristic 2 has exactly $q = 2^h$ geometric q-torsion points, it follows that

$$B = E[q](k),$$

and the descended action of C is exactly translation by $E[q](k)$.

Choose a nonzero invariant differential ω on E. The double cover $Y \to E$ is branched exactly over $E[q](k)$, with lower jump 1 at every branch point. Moreover, C commutes with u and $C \cap \langle u \rangle = 1$. Lemma 5.4 therefore gives a unique $\lambda \in k^\times$ such that

$$Y \cong X(E, \omega, \lambda, q).$$

Putting $\eta = \lambda^{-1}\omega$ simply normalizes all residues to 1, and hence

$$Y \cong X(E, \eta, 1, q).$$

If $h \geq 3$, Proposition 3.1 gives

$$\mathrm{Aut}(Y) \cong \mathrm{Dih}(C_q) \times C_2$$

of order $4q$. Since $|S| = 4q$, we obtain $S = \mathrm{Aut}(Y)$.

It remains to consider $h = 2$. The construction itself gives a subgroup

$$S_0 \cong \mathrm{Dih}(C_4) \times C_2$$

of $\mathrm{Aut}(Y)$ of order 16. Since Y is ordinary of genus 5, Nakajima's bound, applied to an arbitrary 2-subgroup of $\mathrm{Aut}(Y)$, shows that every such subgroup has order at most $4(\gamma(Y) - 1) = 16$. Thus both S and S_0 are Sylow 2-subgroups of $\mathrm{Aut}(Y)$. They are conjugate, and in particular $S \cong \mathrm{Dih}(C_4) \times C_2$. □

Corollary 5.7. *Let (Y, S) be in case* (ib) *and put $q = g(Y) - 1$. Then*

$$S \cong \mathrm{Dih}(C_q) \times C_2.$$

In particular, the other abstract group compatible with the structural description in Theorem 1.1, namely

$$\left\langle c, u, w \;\middle|\; c^q = u^2 = w^2 = 1,\ [c, u] = [u, w] = 1,\ wcw = c^{-1}u \right\rangle,$$

never occurs in case (ib).

Proof. By Proposition 1.4, one has $q = 2^h$ with $h \geq 2$. For $h \geq 3$ this follows from Theorem 5.6, which gives $S = \mathrm{Aut}(Y) \cong \mathrm{Dih}(C_q) \times C_2$. For $h = 2$, the same theorem shows directly that $S \cong \mathrm{Dih}(C_4) \times C_2$.

For completeness, the alternative displayed in the statement is genuinely different from the split group. If $A = C_q \times \langle u \rangle$ and c generates C_q, an involutory automorphism of A fixing u and inducing inversion on $A/\langle u \rangle$ sends c either to c^{-1} or to $c^{-1}u$. These give the split and twisted presentations, respectively. In the twisted case

$$(c^i u^j w)^2 = u^i.$$

Hence there are q involutions outside A, whereas $\mathrm{Dih}(C_q) \times C_2$ has $2q$ involutions outside A. The two groups are therefore not isomorphic. □

Proposition 5.8 (The examples of Giulietti–Korchmáros)**.**

(a) Genus 9. *Let $\mu \in \mathbb{F}_{16}$ satisfy $\mu^4 + \mu + 1 = 0$, and let $\mathcal{X}$ be the smooth projective model of the plane curve of [6, Section 6.2],*

$$(x^7 + \mu^5 x^5 + \mu^{13} x^3 + \mu^9 x)(z^4 + z) = x^8 + \mu^2 x^6 + \mu^8 x^4 + \mu^3 x^2 + \mu^2. \tag{7}$$

Let E_μ be the elliptic curve $y^2 + xy = x^3 + \mu$. Then

$$\mathcal{X} \cong X(E_\mu, dx/x, 1, 8).$$

In particular $\mathcal{X}$ is ordinary of genus 9, and $\mathrm{Aut}(\mathcal{X}) \cong \mathrm{Dih}(C_8) \times C_2$. The pair $(\mathcal{X}, \mathrm{Aut}(\mathcal{X}))$ is of type (ib)*. In the parametrization of Section 5 it corresponds to $(A, c) = (\mu, 1)$.*

(b) Genus 5. *Let $\mathcal{Y}$ be the Artin–Mumford curve*

$$(x^4+x)(y^4+y)=1,$$

used in [6, Section 6.4]; for background on the Artin–Mumford family see also [3]. For $\alpha,\beta\in\mathbb{F}_4$ let $\varphi_{\alpha,\beta}(x,y)=(x+\alpha,y+\beta)$, and let $\rho(x,y)=(y,x)$. Put

$$S=\{\varphi_{\alpha,\beta}\}\rtimes\langle\rho\rangle,\qquad u=\varphi_{1,1},\qquad \bar{\mathcal{Y}}=\mathcal{Y}/\langle u\rangle,\qquad \bar{S}=S/\langle u\rangle.$$

Then $(\bar{\mathcal{Y}},\bar{S})$ is of type (ib)*, and $\bar{\mathcal{Y}}\cong X(E,\eta,1,4)$ for some pair (E,η). The elliptic curve E is the smooth curve $(s^2+s)(t^2+t)=1$ of bidegree $(2,2)$ in $\mathbf{P}^1\times\mathbf{P}^1$.*

Proof. **(a)** Throughout we use the following identities in $\mathbb{F}_{16}$:

$$\mu^4=\mu+1,\quad \mu^5=\mu^2+\mu,\quad \mu^8=\mu^2+1,\quad \mu^9=\mu^3+\mu,\quad \mu^{10}=\mu^2+\mu+1,$$

$$\mu^{13}=\mu^3+\mu^2+1,\quad \mu^{14}=\mu^3+1,\quad \mu^{15}=1.$$

Since the Frobenius map $a\mapsto a^2$ is bijective on $\mathbb{F}_{16}$, we have $\sqrt{\mu}=\mu^8$, $\mu^{1/4}=\mu^4$ and $\mu^{1/8}=\mu^2$.

Step 1: the x-coordinates of $E_\mu[8]$. Take $E_A: y^2+xy=x^3+A$ with $A\in k^\times$. The tangent at $R=(x,y)$ has slope $x+y/x$. Using $(y/x)^2+y/x=x+A/x^2$ we get

$$x(2R)=\frac{x^4+A}{x^2}.$$

The nonzero 2-torsion point is $T_2=(0,\sqrt{A})$.

A point R has order 4 iff $x(2R)=0$, i.e. iff $x^4=A$, i.e. iff $x=A^{1/4}$.

A point R has order 8 iff $x(2R)=A^{1/4}$, i.e. iff

$$x^4+A^{1/4}x^2+A=(x^2+A^{1/8}x+A^{1/2})^2=0.$$

Hence the x-coordinates of the six points of $E_A[8](k)\setminus\{O,T_2\}$ are exactly the roots of

$$P_A(x)=(x+A^{1/4})(x^2+A^{1/8}x+A^{1/2}).$$

These are three distinct nonzero roots. The quadratic factor is separable, and substituting $x=A^{1/4}$ into it gives $A^{3/8}\neq 0$. Each root carries two points of E_A.

For $A=\mu$ this gives

$$P:=P_\mu=x^3+\mu^{10}x^2+\mu^{14}x+\mu^{12},\qquad P'=x^2+\mu^{14}.$$

Squaring coefficient by coefficient,

$$xP^2=x^7+\mu^5x^5+\mu^{13}x^3+\mu^9x.$$

This is the left-hand coefficient in (7).

Step 2: two polynomial identities. Put

$$Q=x^4+\mu x^3+\mu^4x^2+\mu^9x+\mu,\qquad M=\mu x^3+\mu^3x^2+\mu^2x+\mu^5.$$

Then $Q^2=x^8+\mu^2x^6+\mu^8x^4+\mu^3x^2+\mu^2$, which is the right-hand side of (7). We claim

$$M^2+xPM=(x^3+\mu)P^2+xQ^2, \tag{8}$$

$$M+x^2P'=(x+\mu^8)\,P. \tag{9}$$

Both sides of (8) expand to

$$\mu x^7+\mu x^6+\mu^3x^5+\mu^6x^4+\mu x^3+\mu^{14}x^2+\mu^2x+\mu^{10}.$$

The terms x^9 cancel on the right. Identity (9) is a direct expansion.

Two consequences will be used below. By (9), $M(a) = a^2P'(a) \neq 0$ for every root a of P. Moreover $M(0) = \mu^5 = \mu^8 \cdot \mu^{12} = \sqrt{\mu}\, P(0)$.

Step 3: an Artin–Schreier function of the construction. On E_μ put

$$f = \frac{y}{x} + \frac{M(x)}{xP(x)}, \qquad \omega = \frac{dx}{x}.$$

We check the poles and residues of $f\omega$ at each point.

Points over a root a of P. Here $x - a$ is a local parameter, and y/x is regular. The term $M/(xP)$ has a simple pole, and

$$\operatorname{Res}(f\omega) = \frac{M(a)}{a^2P'(a)} = 1.$$

The point $T_2 = (0, \mu^8)$. Put $s = y + \mu^8$. The equation becomes $s^2 = x(s + \mu^8 + x^2)$. Hence s is a local parameter, $\operatorname{ord}(x) = 2$, and

$$\omega = \frac{ds}{\mu^8 + s + x^2}, \qquad \frac{1}{x} = \frac{\mu^8 + s + x^2}{s^2}.$$

Since $M(0)/P(0) = \mu^8$, we can write $y + M/P = s + x\, r(x)$ with r regular at $x = 0$. Therefore

$$f\omega = \frac{s + x\, r(x)}{s^2}\, ds = \Big(\frac{1}{s} + \frac{r(x)}{\mu^8 + s + x^2}\Big) ds.$$

So f has a simple pole at T_2, and the residue is 1.

The point O. Here y/x has a simple pole, and $M/(xP)$ is regular because $\deg M = 3 < 4$. The residue theorem then gives $\operatorname{Res}_O(f\omega) = 7 \cdot 1 = 1$.

All other points. f is regular there.

By Step 1, f has simple poles exactly on $E_\mu[8](k)$, with all residues equal to 1 with respect to dx/x. By the independence of the choice of f (proof of Theorem 1.7), the cover $z^2 + z = f$ is $X(E_\mu, dx/x, 1, 8)$.

Step 4: the plane model. Put $W = z^2 + z = f$. Then

$$y = xW + M/P.$$

Substitute this into $y^2 + xy = x^3 + \mu$ and use (8):

$$x^2(W^2 + W) = x^3 + \mu + \frac{M^2 + xPM}{P^2} = \frac{xQ^2}{P^2}.$$

Since $W^2 + W = z^4 + z$, this is exactly $xP^2(z^4 + z) = Q^2$, i.e. (7).

Conversely, $y \in k(x, z)$, so $k(x, z) = k(E_\mu)(z)$. This field has degree 4 over $k(x)$. Hence (7) is irreducible, and its smooth model is $X(E_\mu, dx/x, 1, 8)$. Theorem 1.7 now gives all remaining assertions.

(b)

Genus and smoothness. The affine part of $\mathcal{Y}$ is smooth. Indeed, $\partial_x = y^4 + y$ and $\partial_y = x^4 + x$ vanish simultaneously only where $(x^4 + x)(y^4 + y) = 0 \neq 1$. At $X_\infty = (1 : 0 : 0)$ the tangent cone is $\prod_{\nu \in \mathbb{F}_4}(Y + \nu Z)$, so X_∞ is an ordinary 4-fold point. Its four branches correspond to the tangents $Y = \nu Z$. The same holds at Y_∞ with the tangents $X = \nu Z$. Hence $g(\mathcal{Y}) = 21 - 2 \cdot 6 = 9$.

The action on points.

- $\varphi_{\alpha,\beta}$ moves every affine point unless $\alpha = \beta = 0$.

- At X_∞ it sends the branch with tangent $Y = \nu Z$ to the branch with tangent $Y = (\nu+\beta)Z$.
- At Y_∞ it sends the branch with tangent $X = \nu Z$ to the branch with tangent $X = (\nu+\alpha)Z$.
- ρ interchanges X_∞ and Y_∞.

u acts freely. In particular $u = \varphi_{1,1}$ has no fixed point, and Riemann–Hurwitz gives $g(\bar{\mathcal{Y}}) = 5$.

Orbits of S and of $\bar{S}$. No nontrivial translation $\varphi_{\alpha,\beta}$ fixes an affine point. If two distinct elements of the coset $S \setminus \{\varphi_{\alpha,\beta} : \alpha, \beta \in \mathbb{F}_4\}$ fixed the same affine point, their quotient would be a nontrivial translation fixing that point. Thus an affine stabilizer has order at most 2, and every S-orbit on affine points has at least 16 elements. The eight branches at infinity form a single S-orbit. Since u acts freely, their images form a $\bar{S}$-orbit of size 4, while every other $\bar{S}$-orbit has size at least 8. In particular, every $\bar{S}$-orbit has at least 4 points.

Hypotheses (I) and (II). From the orbit count, $\bar{S}$ satisfies (II). It also satisfies (I), since $|\bar{S}| = 16 \geq 8$ and $16 > 2(5-1)$.

A non-inductive central involution. Let ψ be the image of $\varphi_{0,1}$ in $\bar{S}$, which equals the image of $\varphi_{1,0} = \varphi_{0,1}u$. It commutes with the image of $\{\varphi_{\alpha,\beta}\}$. Since $\rho\varphi_{0,1}\rho = \varphi_{1,0}$, the element ψ is a central involution of $\bar{S}$.

Its quotient is $\bar{\mathcal{Y}}/\langle\psi\rangle = \mathcal{Y}/V$ with $V = \{\varphi_{\alpha,\beta} : \alpha, \beta \in \mathbb{F}_2\}$, a group of order 4. The functions $s = x^2 + x$ and $t = y^2 + y$ are V-invariant and satisfy $(s^2 + s)(t^2 + t) = 1$. Moreover $[k(x, y) : k(s, t)] \leq 4 = |V|$, so $k(s, t) = k(x, y)^V$.

The curve $(s^2 + s)(t^2 + t) = 1$ of bidegree $(2, 2)$ in $\mathbf{P}^1 \times \mathbf{P}^1$ is smooth. In the affine chart, the partial derivatives are $t^2 + t$ and $s^2 + s$. In the chart $s = 1/\sigma$ the partial derivative in t at $\sigma = 0$ equals 1. The point (∞, ∞) is not on the curve. Hence the curve has genus 1, and ψ is not inductive.

Determining the case. $\bar{S}$ contains the elementary abelian group $\{\varphi_{\alpha,\beta}\}/\langle u\rangle \cong C_2^3$. A dihedral group of order 16 has no such subgroup, so $\bar{S}$ is not dihedral. Also $|\bar{S}| = 16 \neq 2g + 2 = 12$. Because ψ is not inductive, case (iii) of Theorem 1.1 is excluded. Case (ii) is excluded by the order, and case (ia) by non-dihedrality. Hence $(\bar{\mathcal{Y}}, \bar{S})$ is of type (ib).

Conclusion. Theorem 5.6 with $h = 2$ gives $\bar{\mathcal{Y}} \cong X(E, \eta, 1, 4)$. The elliptic curve E is the quotient by the non-inductive central involution, namely ψ. Hence E is the curve $(s^2+s)(t^2+t) = 1$. □

Remark 5.9. We record four points about the published version of [6].

(1) The displayed equation in Section 6.2 lacks a "+" between YX^7 and $\mu^5 YX^5$. Equation (7) is the evident reading.

(2) Section 6.2 specifies only that μ is a primitive element of $\mathbb{F}_{16}$. In Proposition 5.8(a) we make the field presentation explicit by imposing $\mu^4 + \mu + 1 = 0$. The identification proved there is for this convention; no assertion about a different choice of primitive element is needed for the argument.

(3) At the end of Section 6.2 it is stated that $\bar{\mathcal{X}} = \mathcal{X}/C_2$ has genus 5, has a dihedral subgroup of order 8, and is therefore of type (ib). That inference cannot be made from the displayed subgroup: a group of order 8 on a genus-5 curve does not satisfy the strict inequality in hypothesis (I), and an acting group in case (ib) would have order $4(g - 1) = 16$. The preceding MAGMA computation instead gives the genus-9 curve $\mathcal{X}$ a subgroup $D_8 \times C_2$ of order 32, which is the case-(ib) curve identified in Proposition 5.8(a).

(4) The statement in Section 6.4 that no nontrivial element of S fixes a point of $\mathcal{Y}$ is not correct. The involution $\rho : (x, y) \mapsto (y, x)$ fixes the four affine points (a, a) with $a^4 + a + 1 = 0$, and $\varphi_{\alpha,0}$ with $\alpha \neq 0$ fixes the four branches at X_∞. This does not affect the use made here of that example. In particular, [6, Lemma 3.2] states that under (I) and (II) the

action has two short orbits; what is required is that S have no *global* fixed point, not that every nontrivial element act fixed-point-freely. The orbit computation in the proof of Proposition 5.8(b) verifies hypothesis (II) directly for both S and $\bar{S}$.

5.3 The parameter space

Theorem 5.10. *Fix $q \geq 8$. For*

$$E_A : \qquad y^2 + xy = x^3 + A, \qquad A \in k^\times,$$

the assignment

$$(A, c) \longmapsto X(E_A, c\,dx/x, 1, q)$$

induces a bijection

$$(k^\times)^2 \xrightarrow{\sim} \{\textit{isomorphism classes of type-}(\mathrm{ib})\ \textit{curves of genus } q+1\}.$$

Proof. By Proposition 5.1 and Theorem 5.6, no additional parameter occurs. We recall briefly the normal form for an ordinary elliptic curve in characteristic 2. Since $j(E) \neq 0$, a Weierstrass equation may be normalized to

$$y^2 + xy = x^3 + a_2 x^2 + a_6, \qquad a_6 \neq 0.$$

Replacing y by $y + rx$, where $r^2 + r = a_2$ (possible because k is algebraically closed), removes the x^2-term. Thus

$$E \cong E_A : \quad y^2 + xy = x^3 + A, \qquad A = a_6 \in k^\times,$$

and for this model $j(E_A) = A^{-1}$. Hence A is determined by the j-invariant. Since $H^0(E_A, \Omega^1_{E_A})$ is one-dimensional and dx/x is a nonzero invariant differential, every pair (E, η) is represented by

$$(E_A, c\,dx/x), \qquad A, c \in k^\times.$$

For fixed A, Theorem III.10.1 of [8] gives $\mathrm{Aut}(E_A, 0) = \{1, [-1]\}$, and in characteristic 2 both automorphisms fix dx/x. Hence two pairs $(E_A, c\,dx/x)$ and $(E_B, d\,dx/x)$ are isomorphic if and only if $A = B$ and $c = d$. The assertion follows. □

Remark 5.11. For $q = 4$, Theorem 5.6 still gives surjectivity of the same construction onto the type-(ib) curves of genus 5. However, Proposition 5.1 is stated only for $q \geq 8$, because its proof uses uniqueness of the bielliptic involution. We therefore do not claim that the parameters (A, c) are unique in genus 5.

5.4 The unramified cyclic tower

Proposition 5.12. *Let $C = \langle \widetilde{\tau} \rangle \cong C_q$. For $0 \leq r \leq h$, let $C_r \leq C$ be the unique subgroup of order 2^r and put*

$$X_r = X/C_r.$$

Let $K_r \leq T = E[2^h](k)$ be the image of C_r under the projection $C \to T$, and put

$$E_r = E/K_r, \qquad q_r = 2^{h-r}.$$

Then:

(i) *C_r acts freely on X;*

(ii) *$X \to X_r$ is an unramified cyclic cover of degree 2^r;*

(iii) *X_r is ordinary and*

$$g(X_r) = 1 + q_r = 1 + 2^{h-r};$$

(iv) *in particular, $X/C = X_h$ is an ordinary curve of genus 2;*

(v) *for $0 \le r \le h-1$, the quotient by the image of the deck involution is*

$$X_r/\langle \bar{u} \rangle \cong E_r,$$

and this Artin–Schreier double cover is ramified exactly over

$$E_r[2^{h-r}](k),$$

with lower jump 1 at every branch point;

(vi) *for $0 \le r \le h-1$, choose the unique invariant differential ω_r on E_r satisfying $\phi_r^* \omega_r = \omega$, where $\phi_r : E \to E_r$ is the quotient isogeny. Then X_r is obtained by the same residue construction with data*

$$(E_r, \omega_r, \lambda, q_r).$$

In particular, if $0 \le r \le h-3$, then

$$\mathrm{Aut}(X_r) \cong \mathrm{Dih}(C_{q_r}) \times C_2, \qquad |\mathrm{Aut}(X_r)| = 4q_r = 4(g(X_r)-1),$$

and X_r is again of type (ib).

Thus the unramified quotient tower remains inside the same non-dihedral extremal family from genus $2^h + 1$ down to genus 9.

Proof. Every nontrivial element of C projects to a nontrivial translation of E, hence has no fixed point on X. Thus every C_r acts freely, and $X \to X_r$ is unramified. Riemann–Hurwitz yields

$$2g(X) - 2 = 2^r\big(2g(X_r) - 2\big),$$

so

$$g(X_r) - 1 = 2^{h-r} = q_r.$$

For the 2-rank, the Deuring–Shafarevich formula for the free C_r-action gives

$$\gamma(X) - 1 = 2^r\big(\gamma(X_r) - 1\big).$$

Since $\gamma(X) - 1 = 2^h$, we obtain

$$\gamma(X_r) = 1 + 2^{h-r} = g(X_r),$$

and hence every X_r is ordinary.

Because C_r commutes with u, the involution u descends to X_r and

$$X_r/\langle \bar{u} \rangle \cong X/\langle C_r, u \rangle \cong E/K_r = E_r.$$

The isogeny $\phi_r : E \to E_r$ is the quotient by a free translation subgroup, hence is finite *étale*. Moreover, $C_r \cap \langle u \rangle = 1$, so X is the normalization of the fibre product

$$X_r \times_{E_r} E.$$

Consequently the completed local extension for $X_r \to E_r$ becomes the completed local extension for $X \to E$ after the unramified base change $E \to E_r$. Ramification groups and lower jumps are therefore unchanged. The branch locus is the image of T in E_r, a cyclic group of order q_r.

The elliptic curve E_r is ordinary, being isogenous to E, and therefore this image is precisely $E_r[2^{h-r}](k)$. Thus every branch point has lower jump 1.

Assume now $r \leq h-1$, so q_r is even. Since ϕ_r is separable, pullback on invariant differentials is an isomorphism; hence there is a unique nonzero invariant differential ω_r on E_r such that

$$\phi_r^* \omega_r = \omega.$$

The cyclic group $C/C_r \cong C_{q_r}$ acts on X_r, commutes with the image of the deck involution, has trivial intersection with it, and maps isomorphically onto the translation subgroup $E_r[q_r](k)$. Together with the branch and jump computation above, Lemma 5.4 applies and yields a unique $\lambda_r \in k^\times$ such that

$$X_r \cong X(E_r, \omega_r, \lambda_r, q_r).$$

It remains to identify λ_r. The curve X is the normalization of $X_r \times_{E_r} E$, and the map $\phi_r : E \to E_r$ is *étale*. Let $P \in E$ lie above $P_r \in E_r$. For a local Artin–Schreier representative f_r of X_r/E_r, the map on completed local fields induced by the finite *étale* morphism ϕ_r has ramification index 1. Hence pullback preserves residues, and

$$\mathrm{Res}_P\big(\phi_r^*(f_r \omega_r)\big) = \mathrm{Res}_{P_r}(f_r \omega_r).$$

Since $\phi_r^* \omega_r = \omega$ and the pullback cover is X/E, the left-hand side equals λ. Hence

$$\lambda_r = \lambda.$$

Thus

$$X_r \cong X(E_r, \omega_r, \lambda, q_r).$$

If $r \leq h-3$, then $q_r \geq 8$, so Theorem 1.7 applies to this new set of data. This gives

$$\mathrm{Aut}(X_r) \cong \mathrm{Dih}(C_{q_r}) \times C_2$$

and shows that X_r is again a non-dihedral Nakajima extremal curve of type (ib). □

6 Central involutions and their quotients

The quotients by the central involutions make the inductive structure of case (ib) explicit and connect it with the dihedral case.

6.1 The centre

Corollary 6.1. *For $q \geq 8$,*

$$Z(\mathrm{Aut}(X)) = \langle \widetilde{\tau}^{q/2}, u \rangle \cong C_2 \times C_2.$$

Among the three nontrivial central involutions, u is the unique one whose quotient has genus 1. The other two act freely, and each has an ordinary quotient of genus $q/2+1$.

Proof. The center of $\mathrm{Dih}(C_q)$ is generated by the unique element of order 2 in C_q, namely $\widetilde{\tau}^{q/2}$. The description of the center follows from Proposition 3.1. The involution u has quotient E of genus one. Both $\widetilde{\tau}^{q/2}$ and $u\widetilde{\tau}^{q/2}$ project to the nonzero 2-torsion translation on E, so they have no fixed point. Riemann–Hurwitz and Deuring–Shafarevich for either free involution give genus and 2-rank $q/2+1$ for the quotient. □

Theorem 6.2. *Assume $q = 2^h$ with $h \geq 4$, put*

$$t = \widetilde{\tau}, \qquad a = t^{q/2}, \qquad V = \langle a, u \rangle = Z(\mathrm{Aut}(X)),$$

and set

$$X^- := X/\langle a \rangle, \qquad X^+ := X/\langle au \rangle, \qquad E_1 := X/V.$$

Then:

(i) *E_1 is an ordinary elliptic curve and the natural map $E = X/\langle u \rangle \to E_1$ is a separable isogeny of degree 2;*

(ii) *X^- is an ordinary Nakajima extremal curve of genus*

$$g(X^-) = \frac{q}{2} + 1$$

whose full automorphism group is

$$\mathrm{Aut}(X^-) \cong \mathrm{Dih}(C_{q/2}) \times C_2.$$

Moreover, for any nonzero invariant differential ω_1 on E_1, there is a $\lambda_1 \in k^\times$ such that

$$X^- \cong X(E_1, \omega_1, \lambda_1, q/2).$$

In particular X^- is again a member of the constructed family and is of type (ib)*;*

(iii) *X^+ is an ordinary Nakajima extremal curve of the same genus, but*

$$\mathrm{Aut}(X^+) \cong \mathrm{Dih}(C_q);$$

in particular X^+ is of the dihedral type (ia)*;*

(iv) *relative to the pair (X, G), the involutions a and au are exactly the two central inductive involutions in the sense of [6]; the bielliptic involution u is not inductive.*

Proof. By Corollary 6.1, both a and au act freely on X. Hence Riemann–Hurwitz and Deuring–Shafarevich give

$$g(X^-) = g(X^+) = \frac{q}{2} + 1, \qquad \gamma(X^-) = \gamma(X^+) = \frac{q}{2} + 1.$$

Thus both quotients are ordinary. Moreover,

$$E_1 = X/\langle a, u \rangle \cong E/\langle \bar{a} \rangle,$$

where $\bar{a}$ is translation by the unique nonzero 2-torsion point of E. The map $E \to E_1$ is therefore the separable degree-2 isogeny with kernel $\langle \bar{a} \rangle$, and E_1 is ordinary. Put

$$T_1 := E_1[2^{h-1}](k).$$

Consider first X^-. The group $G/\langle a \rangle$ acts faithfully on X^- and

$$G/\langle a \rangle \cong \mathrm{Dih}(C_{q/2}) \times C_2,$$

of order $2q = 4(g(X^-) - 1)$. The image of u is central and

$$X^-/\langle \bar{u} \rangle = E_1.$$

The subgroup $\langle a\rangle$ is the unique subgroup of order 2 in $C = \langle t\rangle$. Thus $X^- = X/\langle a\rangle$ is precisely the case $r = 1$ of Proposition 5.12, and $E_1 = E/\langle \bar{a}\rangle$ is the corresponding elliptic quotient. Let ω_1^0 be the unique invariant differential on E_1 satisfying

$$(E \to E_1)^*\omega_1^0 = \omega.$$

Proposition 5.12(vi) gives directly

$$X^- \cong X(E_1, \omega_1^0, \lambda, q/2).$$

If ω_1 is any other nonzero invariant differential on E_1, write $\omega_1 = c\omega_1^0$ with $c \in k^\times$. The defining residue condition then gives

$$X(E_1, \omega_1^0, \lambda, q/2) \cong X(E_1, \omega_1, c\lambda, q/2).$$

Thus the assertion in part (ii) holds with $\lambda_1 = c\lambda$.

Since $h \geq 4$, one has $q/2 = 2^{h-1} \geq 8$. The displayed identification realizes X^- as a curve of Theorem 1.7 with parameter $q/2$. Thus Theorem 1.7(iii)–(iv) gives directly

$$\mathrm{Aut}(X^-) \cong \mathrm{Dih}(C_{q/2}) \times C_2,$$

and shows that X^- is of type (ib).

Now put $H = \langle au\rangle$. In G/H the images of a and u coincide, while the image of t still has order q. Hence

$$G/H \cong \mathrm{Dih}(C_q),$$

of order $2q = 4(g(X^+) - 1)$. The common image $\bar{a} = \bar{u}$ is a bielliptic involution and

$$X^+/\langle \bar{u}\rangle = E_1.$$

We claim that the branch locus is again exactly T_1. A point $\overline{P} \in X^+ = X/H$ is fixed by the common image $\bar{u} = \bar{a}$ precisely when, for a lift $P \in X$, one has either $uP = P$ or $aP = P$: indeed uP must lie in the H-orbit of P, and $H = \{1, au\}$. The second alternative is impossible because a acts freely. Thus the fixed points of $\bar{u}$ are precisely the images of the u-fixed points on X. Their images in E_1 are the image of T under the degree-2 isogeny $E \to E_1$, which is precisely T_1. Hence this is exactly the branch locus of $X^+ \to E_1$.

Since $h \geq 4$, one has $g(X^+) = q/2 + 1 \geq 9$, so the bielliptic involution of X^+ is unique by Castelnuovo–Severi. By uniqueness of the bielliptic involution, every automorphism descends to (E_1, T_1). Here $|T_1| = q/2$, and, as in Proposition 3.1,

$$\mathrm{Aut}(E_1, T_1) = T_1 \rtimes \langle [-1]\rangle$$

has order q. Thus $|\mathrm{Aut}(X^+)| \leq 2q$, and the dihedral subgroup G/H already has this order. Therefore

$$\mathrm{Aut}(X^+) \cong \mathrm{Dih}(C_q).$$

The image of t on E_1 is a nontrivial translation, so neither of the two displayed extremal groups has a global fixed point.

For X^+ the acting group is dihedral of order $4(g(X^+) - 1) = 2q$. Case (ii) would require $2q = 2g(X^+) + 2 = q + 4$, hence $q = 4$, whereas here $h \geq 4$. Moreover $Z(\mathrm{Dih}(C_q)) = \langle \bar{u}\rangle$, and $X^+/\langle \bar{u}\rangle = E_1$ has genus 1. Thus $\bar{u}$ is non-inductive, so case (iii) is also excluded. The classification therefore places X^+ in case (ia). Moreover,

$$|G/\langle a\rangle| = |G/\langle au\rangle| = 2q = 4\left(\frac{q}{2} + 1 - 1\right),$$

and both induced groups have order at least 8 and no global fixed point. Thus a and au are inductive central involutions. On the other hand, $X/\langle u\rangle = E$ has genus one, so u is not inductive. Since $Z(G) = \langle a, u\rangle$ has exactly three nontrivial elements, a and au are precisely the two central inductive involutions.

□

Remark 6.3. When $h = 3$ (so $q = 8$), the quotient $X^- = X/\langle a \rangle$ has genus 5. Proposition 5.12(vi) identifies it with a curve of the form $X(E_1, \omega_1, \lambda_1, 4)$, and Remark 7.3 shows that it is still of type (ib). In this genus we do not claim a determination of the full automorphism group.

7 Explicit equations and a logarithmic-differential realization

The construction admits the following explicit form. The proposition is stated for residue 1; multiplication by $\lambda \in k^\times$ gives the general case.

Proposition 7.1. *Let $q = 2^h$ and let E/k be an ordinary elliptic curve. Choose $R \in E(k)$ of order $2q$ and put*

$$T = \langle 2R \rangle.$$

Then there is $H \in k(E)^\times$ with

$$\operatorname{div}(H) = \sum_{P \in T} [P] - q[R].$$

If ω is a nonzero invariant differential, then

$$f = \frac{dH/H}{\omega}$$

has a simple pole at every point of T and no other pole, and

$$\operatorname{Res}_P(f\omega) = 1 \qquad (P \in T).$$

Proof. The subgroup T is cyclic of even order q, so the sum of its elements in the group law of E is its unique nonzero point of order 2, namely qR. Therefore the degree-zero divisor

$$\sum_{P \in T} [P] - q[R]$$

has trivial class in $\mathrm{Pic}^0(E)$ and is principal.

At a zero $P \in T$ of H, the logarithmic differential dH/H has a simple pole with residue 1. At R, write locally $H = t^{-q} v$ with v a unit. Then

$$\frac{dH}{H} = -q\frac{dt}{t} + \frac{dv}{v} = \frac{dv}{v},$$

because $q = 0$ in characteristic 2. Hence dH/H is regular at R. Dividing by the nowhere-vanishing differential ω proves the claim. □

Proposition 7.2 (A genus-9 equation over $\mathbb{F}_2$)**.** *Let $k = \overline{\mathbb{F}}_2$ and consider the ordinary elliptic curve*

$$E: \qquad y^2 + xy = x^3 + 1.$$

Let $\rho \in \mathbb{F}_4 \setminus \mathbb{F}_2$ satisfy $\rho^2 + \rho + 1 = 0$. Then $P_0 = (\rho, 0)$ has order 8, and

$$T = E[8](k) = E(\mathbb{F}_4) = \{O, (0,1)\} \cup \{(a,0), (a,a) : a \in \mathbb{F}_4^\times\}.$$

With the invariant differential $\omega = dx/x$, put

$$f = \frac{y}{x} + \frac{x^4 + x^3 + 1}{x^4 + x}. \tag{10}$$

Then f has simple poles precisely at the points of T and

$$\operatorname{Res}_P(f\omega) = 1 \qquad (P \in T).$$

Consequently the smooth projective model of

$$z^2 + z = \frac{y}{x} + \frac{x^4 + x^3 + 1}{x^4 + x}, \qquad y^2 + xy = x^3 + 1, \tag{11}$$

is an ordinary curve of genus 9 *and*

$$\mathrm{Aut}(X) \cong \mathrm{Dih}(C_8) \times C_2.$$

It is birational to the plane curve

$$\boxed{(x^7 + x)(z^4 + z) = x^8 + x^6 + x^4 + x^2 + 1}. \tag{12}$$

*In particular, this genus-*9 *member of the family has a plane model over* $\mathbb{F}_2$.

Proof. The curve E is ordinary, and a direct use of the chord–tangent law gives

$$[2](\rho, 0) = (1, 0), \qquad [4](\rho, 0) = (0, 1).$$

Since $(0, 1)$ is the nontrivial 2-torsion point, P_0 has order 8. The displayed set has eight points and is exactly $\langle P_0 \rangle$.

We verify the principal parts of f. First let $P = (a, b)$ with $a \in \mathbb{F}_4^\times$. Then $a^3 = 1$, the coordinate $x - a$ is a local parameter at P, and the second term of (10) is the only singular term. Since

$$x^4 + x = x(x^3 + 1), \qquad \frac{d}{dx}(x^4 + x) = 1,$$

and $a^4 + a^3 + 1 = a$, one obtains

$$\mathrm{Res}_P(f\omega) = \frac{a}{a} = 1.$$

This accounts for the six points of T with nonzero x-coordinate.

At $P = (0, 1)$ put $s = y + 1$. The equation of E becomes

$$s^2 = x(1 + s + x^2),$$

so s is a local parameter and

$$\omega = \frac{dx}{x} = \frac{ds}{1 + s + x^2}.$$

Writing (10) over the common denominator $x(1 + x^3)$ gives

$$f = \frac{s(1 + x^3) + x^4}{x(1 + x^3)}.$$

Therefore

$$f\omega = \left(\frac{1}{s} + \frac{x^4}{s^2(1 + x^3)} \right) ds.$$

As $\mathrm{ord}_s(x) = 2$, the second summand is regular, and the residue is again 1. Finally, at O one has $\mathrm{ord}_O(x) = -2$ and $\mathrm{ord}_O(y) = -3$, so f has a simple pole there and no further poles occur. The residue theorem, together with the seven finite residues already computed, gives $\mathrm{Res}_O(f\omega) = 1$.

Thus (11) is the case $q = 8$, $\lambda = 1$ of the main construction, and Theorem 1.7 yields the genus, ordinarity, and full automorphism group.

For the plane equation set $W = z^2 + z$. From (11),

$$y = xW + \frac{x^4 + x^3 + 1}{x^3 + 1}.$$

Substitution in $y^2 + xy = x^3 + 1$ and simplification in characteristic 2 gives

$$W^2 + W = \frac{x^8 + x^6 + x^4 + x^2 + 1}{x^7 + x}.$$

Since $W^2 + W = z^4 + z$, this is precisely (12). Conversely, the displayed expression for y shows that the function field defined by (12) contains the function field of (11). The latter has degree 4 over $k(x)$, while (12) has degree 4 in z. Hence the polynomial in z in (12) is irreducible over $k(x)$ and the two function fields coincide. Thus the two models are birational. □

Remark 7.3 (The case $q = 4$). The residue construction also works for $q = 4$ and gives an ordinary curve of genus 5 with a subgroup

$$S \cong \mathrm{Dih}(C_4) \times C_2$$

of order $16 = 4(g - 1)$. The same construction shows that S has no global fixed point, while the central deck involution u has elliptic quotient and hence is not inductive. Moreover S is not dihedral, because its center has order 4 whereas the center of a dihedral group of order 16 has order 2, and $16 \neq 2g + 2 = 12$. Thus cases (ia), (ii), and (iii) of Theorem 1.1 are excluded, so the constructed curve is of type (ib).

Theorem 5.6 shows conversely that every curve of type (ib) of genus 5 arises in this way. Proposition 5.8 provides a direct verification for the genus-5 example arising from [6, Sections 6.4–6.5]. The uniqueness argument for the bielliptic involution does not apply in genus 5, so we do not claim that the parameters (A, c) are unique. Nakajima's bound shows only that the displayed group is a Sylow 2-subgroup of the full automorphism group. We therefore do not extend the assertion on the full automorphism group to $q = 4$.

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Declaration on the use of generative AI

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